Attitude adjustment: Trends in the semantics of clausal-embedding

Deniz Özyıldız, Tom Roberts
Universität Stuttgart, Universiteit Utrecht

ESSLLI 2026 · Univerzita Karlova
Aug. 10–15
Day 5

Yesterday (with corrections)

(1) Jones buttered the toast.
$\exists e : \mathit{butter}(e) \wedge \mathit{agent}(e) = \mathit{jones} \wedge \mathit{theme}(e) = \iota x[\mathit{toast}(x)]$

I'll explain ι in just a second, it's similar to 'the'.

Yesterday (with corrections)

(1) Jones buttered the toast.
$\exists e : butter(e) \wedge agent(e) = jones \wedge theme(e) = \iota x[toast(x)]$

I'll explain ι in just a second, it's similar to 'the'.

(2) Chloe believes the rumor.
$\exists e : believe(e) \wedge experiencer(e) = chloe \wedge theme(e) = \iota x[rumor(x)]$

Yesterday (with corrections)

(1) Jones buttered the toast.
$\exists e : butter(e) \wedge agent(e) = jones \wedge theme(e) = \iota x[toast(x)]$

I'll explain ι in just a second, it's similar to 'the'.

(2) Chloe believes the rumor.
$\exists e : believe(e) \wedge experiencer(e) = chloe \wedge theme(e) = \iota x[rumor(x)]$

(3) Chloe believes the rumor that it's raining.
$\exists e : believe(e) \wedge experiencer(e) = chloe \wedge theme(e) = \iota x[rumor(x) \wedge content(x) = \{w : rain\ at\ w\}]$

Yesterday (with corrections)

(1) Jones buttered the toast.
$\exists e : butter(e) \wedge agent(e) = jones \wedge theme(e) = \iota x[toast(x)]$

I'll explain ι in just a second, it's similar to 'the'.

(2) Chloe believes the rumor.
$\exists e : believe(e) \wedge experiencer(e) = chloe \wedge theme(e) = \iota x[rumor(x)]$

(3) Chloe believes the rumor that it's raining.
$\exists e : believe(e) \wedge experiencer(e) = chloe \wedge theme(e) = \iota x[rumor(x) \wedge content(x) = \{w : rain\ at\ w\}]$

(4) Chloe believes that it's raining.
$\exists e, x : believe(e) \wedge experiencer(e) = chloe \wedge theme(e) = x \wedge content(x) = \{w : rain\ at\ w\}$

Some features of the system from yesterday

- clauses are semantic modifiers of nouns (and later today, of verb phrases)
- modal displacement is anchored to contentful entities
- this shift in thinking might help capture natural language data better

Outline

Questions from yesterday

Today

Bit 1: Content of the theme and the subject's beliefs

"With a statement like (5), don't we have to/how do we link the content of the theme to the belief state of the subject, i.e., to (6)?"

(5) $\exists e, x : \textit{believe}(e) \wedge \textit{experiencer}(e) = \textit{Chloe} \wedge \textit{theme}(e) = x \wedge \textit{content}(x) = \{w : \textit{rain at } w\}$

(6) $\textit{Dox}(\textit{Chloe}) \subseteq \{w : \textit{rain at } w\}$

Bit 1: Content of the theme and the subject's beliefs

"With a statement like (5), don't we have to/how do we link the content of the theme to the belief state of the subject, i.e., to (6)?"

(5) $\exists e, x : believe(e) \wedge experiencer(e) = Chloe \wedge theme(e) = x \wedge content(x) = \{w : rain\ at\ w\}$

(6) $Dox(Chloe) \subseteq \{w : rain\ at\ w\}$

Answer #1:
No. (5) is what it means for x to believe p now. DOX out.

Bit 1: Content of the theme and the subject's beliefs

"With a statement like (5), don't we have to/how do we link the content of the theme to the belief state of the subject, i.e., to (6)?"

(5) $\exists e, x : \mathit{believe}(e) \wedge \mathit{experiencer}(e) = \mathit{Chloe} \wedge \mathit{theme}(e) = x \wedge \mathit{content}(x) = \{w : \mathit{rain\ at\ w}\}$

(6) $\mathit{Dox}(\mathit{Chloe}) \subseteq \{w : \mathit{rain\ at\ w}\}$

Answer #1:
No. (5) is what it means for x to believe p now. DOX out.

Answer #2: Yes, by assuming that (7) must hold (to be refined):

(7) $\forall x, y, e : \mathit{believe}(e) \wedge \mathit{experiencer}(e) = x \wedge \mathit{theme}(e) = y$
$\Leftrightarrow$
$\mathit{DOX}(x) \subseteq \mathit{content}(y)$

Bit 1: Content of the theme and the subject's beliefs

"With a statement like (5), don't we have to/how do we link the content of the theme to the belief state of the subject, i.e., to (6)?"

(5) $\exists e, x : \mathit{believe}(e) \wedge \mathit{experiencer}(e) = \mathit{Chloe} \wedge \mathit{theme}(e) = x \wedge \mathit{content}(x) = \{w : \mathit{rain\ at\ w}\}$

(6) $\mathit{Dox}(\mathit{Chloe}) \subseteq \{w : \mathit{rain\ at\ w}\}$

Answer #1:
No. (5) is what it means for x to believe p now. DOX out.

Answer #2: Yes, by assuming that (7) must hold (to be refined):

(7) $\forall x, y, e : \mathit{believe}(e) \wedge \mathit{experiencer}(e) = x \wedge \mathit{theme}(e) = y \Leftrightarrow \mathit{DOX}(x) \subseteq \mathit{content}(y)$

There's something to think through here.

Bit 2: The denotation of thematic heads

We had subtrees of the form:

$$\{e : \textit{patient}(e) = \textit{the toast}\}$$

the toast   PATIENT

and I didn't tell you how to formalize PATIENT.

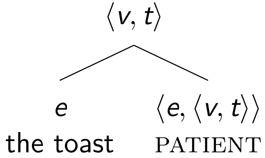
If we agree that 'the toast' is type *e* and that we have to have a set of events at the end, we have the following types, with *e* for entities and *v* for events (new):

$\langle v, t \rangle$

e

the toast PATIENT

If we agree that 'the toast' is type e and that we have to have a set of events at the end, we have the following types, with e for entities and v for events (new):



Now for the function itself:

$$\{e : patient(e) = \iota x[toast(x)]\}$$

$\iota x[toast(x)]$ — the toast

$\lambda x_e.\{e : patient(e) = x\}$ — PATIENT

These two expressions compose via function application, and don't confuse subscript $_e$ with variable e. The first is for 'entitiy', the second for 'event'. For the first time, this is not me being sloppy ;)

To incorporate all of this into the VP, you combine the two middle nodes via set intersection:

$$\{e : butter(e) \wedge patient(e) = \iota x[toast(x)]\}$$

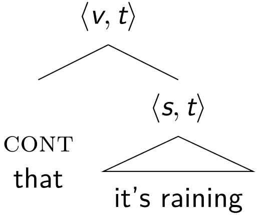
$$\{e : butter(e)\} \qquad \{e : patient(e) = \iota x[toast(x)]\}$$

buttered

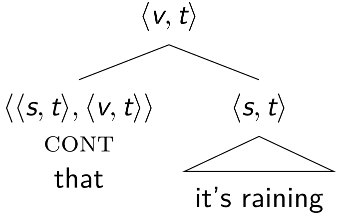
the toast PATIENT

(I'll let you decide on the denotation of, and incorporate the subject "[Jones AGENT]" yourselves)

Similarly for the cont head:



Similarly for the cont head:



$$\{x : \textit{content}(x) = \{w : \textit{rain at } w\}\}$$

$\lambda p_{s,t}.\{x : \textit{content}(x) = p\}$ $\{w : \textit{rain at } w\}$

CONT

that

it's raining

Bit 3: iota

Yesterday we had the statement:

(8) $\textsc{the}[x \text{ s.t. } \mathit{rumor}(x) \wedge \mathit{content}(x) = \{w : \text{it's raining at } w\}]$

and I cited Partee (1986) *Noun phrase interpretation and type-shifting principles*. Here's the correction:

Bit 3: iota

Yesterday we had the statement:

(8) $\textsc{the}[x \text{ s.t. } \mathit{rumor}(x) \wedge \mathit{content}(x) = \{w : \text{it's raining at } w\}]$

and I cited Partee (1986) *Noun phrase interpretation and type-shifting principles*. Here's the correction:

It would have been more accurate for me to use the 'iota' operator ι. It's the simplest formalization of the word 'the'.

(9) a. "rumor that it's raining" type $\langle e, t\rangle$
$\{x : \mathit{rumor}(x) \wedge \mathit{content}(x) = \{w : \text{it's raining at } w\}\}$

b. "the rumor that it's raining" type e
$\iota x[x : \mathit{rumor}(x) \wedge \mathit{content}(x) = \{w : \text{it's raining at } w\}]$

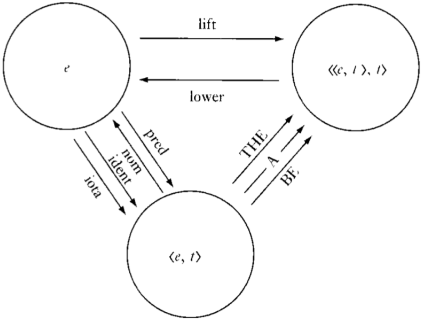


Figure 1

lift	:	$j \rightarrow \lambda P[P(j)]$	total; injective
lower	:	maps a principal ultrafilter onto its generator; lower(lift(j)) = j	partial; surjective
ident	:	$j \rightarrow \lambda x[x = j]$	total; injective
iota	:	$P \rightarrow \iota x[P(x)]$	
		iota(ident(j)) = j	partial; surjective
nom	:	$P \rightarrow {}^{\cap}P$ (Chierchia)	almost total; injective
pred	:	$x \rightarrow {}^{\cup}x$ (Chierchia)	
		pred(nom(P)) = P	partial; surjective

ι vs. THE

Generalized quantifiers like ‘some student’, ‘every student’, ‘three students’, etc. are *not* expressions of type e. (They’re usually treated as being of type $\langle\langle e, t\rangle, t\rangle$)

ι vs. THE

Generalized quantifiers like 'some student', 'every student', 'three students', etc. are *not* expressions of type e. (They're usually treated as being of type $\langle\langle e, t\rangle, t\rangle$)

Yet, you can conjoin generalized quantifiers with individual denoting expressions like 'the teacher':

(10) The teacher and some/every student laughed.

ι vs. THE

Generalized quantifiers like 'some student', 'every student', 'three students', etc. are *not* expressions of type *e*. (They're usually treated as being of type $\langle\langle e, t\rangle, t\rangle$)

Yet, you can conjoin generalized quantifiers with individual denoting expressions like 'the teacher':

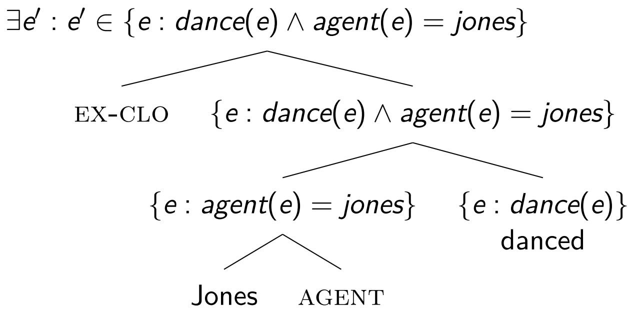
(10) The teacher and some/every student laughed.

Here, we can't use $\iota x[teacher(x)]$, we have to use $\text{THE}(\{x : teacher(x)\})$.

These will yield similar and related meanings, both correspond to the definite article, but have different types.

Bit 4: Existential closure

We said that you could get from a **set** to 'something else' 'through many means'. Another one of these means is **existential closure**.



(11) $[\![\text{EX-CLO}]\!] = \lambda P_{\langle v,t \rangle}.\exists e' : e' \in P$

here, I leave *you* to figure out how to get from $\exists e' \in P$ to $\exists e : X(e) \wedge Y(e) \wedge \ldots$

Bit 4: Existential closure

Existential closure gets us from any set ($\langle \alpha, t \rangle$) to something of type t. We implicitly saw two instances of this yesterday, one along the lines of (11), and one along the lines of (12).

We wanted to get from (12a) to (12b):

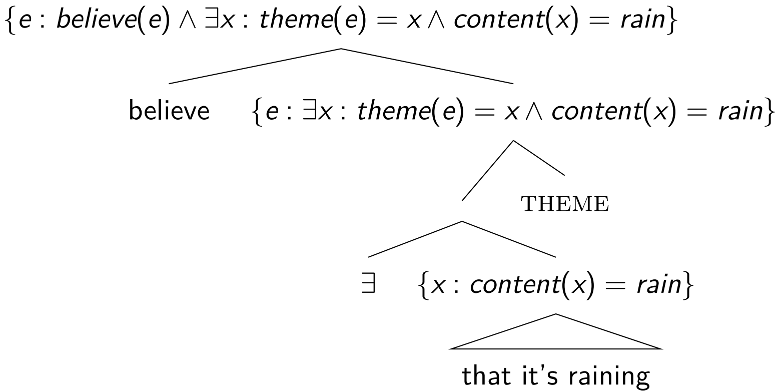
(12) a. 'rumor that it's raining'
$\{x : rumor(x) \wedge content(x) = \{w : rain\ at\ w\}\}$

b. "There's a rumor that it's raining."
$\exists x : rumor(x) \wedge content(x) = \{w : rain\ at\ w\}$

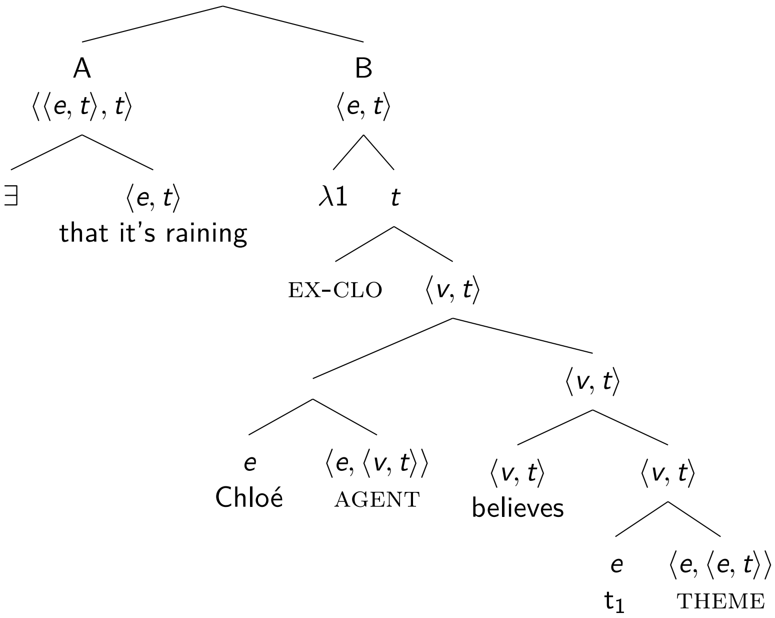
Bit 5: Quantifier Raising

People were justly concerned about the ∃ in the tree below:

- Composition will work out, e.g., via Quantifier Raising would be the most standard way of doing this.
- This problem will disappear later today.



I will not teach quantifier raising today...See Heim & Kratzer (1998: sections 6 & 7).


A
⟨⟨e, t⟩, t⟩
B
⟨e, t⟩
∃
⟨e, t⟩
that it's raining
λ1
t
EX-CLO
⟨v, t⟩
⟨v, t⟩
e
Chloé
⟨e, ⟨v, t⟩⟩
AGENT
⟨v, t⟩
believes
⟨v, t⟩
e
t1
⟨e, ⟨e, t⟩⟩
THEME

The denotations of the two top nodes:

(13) $[\![A]\!] = \lambda P_{\langle\langle e,t\rangle,t\rangle}.\exists x: \mathit{content}(x) = \{w: \mathit{rain\ at\ w}\} \wedge P(x)$

(14) $[\![B]\!] = \lambda x_e.\exists e: \mathit{believe}(e) \wedge \mathit{agent}(e) = \mathit{Chloe} \wedge \mathit{theme}(e) = x$

Outline

Questions from yesterday

Today

An event semantic treatment of attitudes?

We motivated the shift to event semantics based on the observation that ‘believe’ combines with NPs and CPs, and the suspicion that this involves the same ‘believe’.

This doesn’t motivate an *event* semantic treatment per se.

Let’s now make direct use of the event argument.

Substitution failures

Elliott (2020: p. 58)

(55) a. Angela explained [$_{DP}$ the fact that Boris resigned].

explanandum

b. Angela explained [$_{CP}$ that Boris resigned].

explanans

First, we can show that (55a) and (55b) are truth-conditionally independent; (55a) does not entail (55b), and vice versa.

Imagine a context in which everyone is wondering why Boris resigned, and Angela announces that Boris has long-term health issues. Since everyone knows that Boris has in fact resigned, Angela does not bother to mention this. In this scenario, (55a) is *true* but (55b) is *false*.

Now, imagine a context in which, one day Boris does not come in to work, and everyone is wondering why. Angela announces that Boris resigned (which is true), but does not say why. In this context, (55b) is *true* but (55a) is *false*.

Substitution failures

We've illustrated this with 'know the rumor that p' vs. 'know that p', and there are other examples (from Elliott again):

(15) a. Amy remembers the claim that first order logic is undecidable.
b. Amy remembers that first order logic is undecidable.

(16) a. Amy fears the claim that first order logic is undecidable.
b. Amy fears that first order logic is undecidable.

In discussing “the rumor that it’s raining” in yesterday’s system, we’ve discussed the idea that ‘that’ clauses are *modifiers* of NPs.

In discussing “the rumor that it’s raining” in yesterday’s system, we’ve discussed the idea that ‘that’ clauses are *modifiers* of NPs.

The novel idea here is that ‘that’ clauses can also modify VPs.

Entities with propositional content

There are entities with propositional content (rumors, facts, ideas, suggestions, ...), as opposed to ones without (people, chairs, glasses, ...). Things like books, registers, signs, might hold an intermediate place conceptually, but they don't like being given content through a 'that' clause.

(17)
a. the rumor that it's raining
b. *the glasses that it's raining
c. *the sign that we should stop

Events with propositional content

We've talked about beliefs, mostly, and evoked other attitudes on Day 1: saying, wanting, wondering, hoping, … and (in principle) can relate them all to propositions.

Now that we have events at our disposal, we can think of these as *events* that are associated with propositional content, as opposed to ones without: kicking, hugging, jumping, …

We will get ungrammaticality in examples like (18b) and (18c), but it is unclear whether this is due to lexical/syntactic facts or to non-contentfulness.

(18) a. They said that there would be an eclipse.
  b. *They jumped that there would be an eclipse.
  c. *They talked/spoke that there would be an eclipse.

'Talking/speaking' should be no different than 'saying', conceptually, in being associated with content.

If you buy this, then there's a general similarity between 'entities' with content and 'events' with content. This opposes entities and events without content.

Recall our proposal for the set of things with propositional content:

(19) $[\![\text{that it's raining}]\!] = \{x : \textit{content}(x) = \{w : \textit{rain at } w\}\}$

At first we said that x had to range over rumors, ideas, beliefs, facts... What if we also let x range over *events* of believing, knowing, wanting, hoping, etc.?

Let D_c be the set that contains all contentful entities and events, with the exclusion of any non-contentful ones. Then, we more properly write:

(20) $[\![\text{that it's raining}]\!] = \{x \in D_c : \textit{content}(x) = \{w : \textit{rain at } w\}\}$

With this, we can combine ‘that’ clauses with nouns or with verbs.

And this will help explain the explanans vs. explanandum asymmetry.

(21) a. Angela explained the fact that Boris resigned.

b. $\exists e, x : \textit{explain}(e) \wedge \textit{agent}(e) = \textit{boris} \wedge \textit{theme}(x) = \iota y[\textit{fact}(y) \wedge \textit{content}(y) = \{w : \textit{B. resigned at } w\}]$

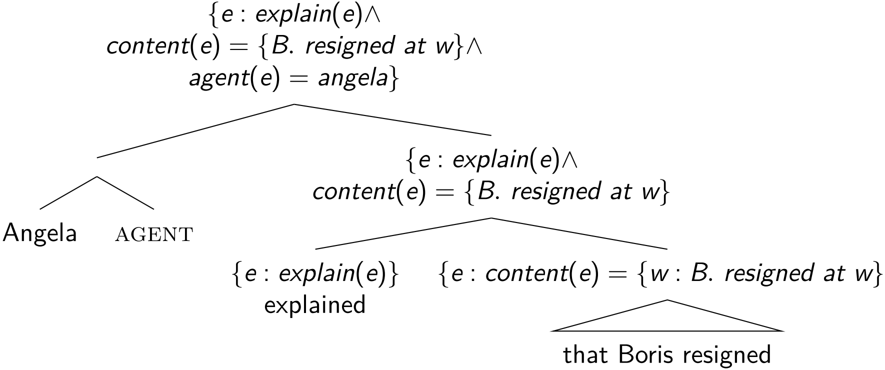
(21) a. Angela explained the fact that Boris resigned.

b. $\exists e, x : explain(e) \wedge agent(e) = boris \wedge theme(x) = \iota y[fact(y) \wedge content(y) = \{w : B.\ resigned\ at\ w\}]$

(22) a. Angela explained that Boris resigned.

b. $\exists e : explain(e) \wedge agent(e) = boris \wedge content(e) = \{w : B.\ resigned\ at\ w\}$

(23) Angela explained that Boris resigned.



This actually solves both the substitution failure problem, but removes the need for extra assumtions in accounting for “believe that p”.

what about believe...

We covered very little of what was in our course description...Nonetheless, I hope that the materials from this week will make reading and doing research in attitude reports more accessible?